

Application: Systematic Sampling Using Set Notation

Consider a population consisting of

$$N = 45,000$$

individuals. A researcher wishes to select a sample of

$$n = 680$$

individuals using **systematic random sampling**.

The sampling interval is determined by

$$k = \left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{45,000}{680} \right\rfloor = 66.$$

A random starting point is selected from the first sampling interval. Suppose the researcher randomly chooses

$$r = 12.$$

The first individual selected is therefore the 12th member of the population. Every 66th individual thereafter is included in the sample. The selected individuals are

$$12, 78, 144, 210, 276, \dots$$

This sequence forms an **arithmetic progression** with

- First term:

$$a_1 = 12,$$

- Common difference:

$$d = 66.$$

Hence, the i -th sampled individual is given by

$$n_i = 12 + 66(i - 1), \quad i = 1, 2, \dots, 680.$$

The complete sample can therefore be expressed using set notation as

$$S = \{12 + 66(i - 1) \mid i = 1, 2, \dots, 680\}.$$

Alternatively, using universal quantification,

$$\forall i \in \{1, 2, \dots, 680\}, \quad n_i = 12 + 66(i - 1).$$

The final sampled individual is

$$n_{680} = 12 + 66(679) = 44,826,$$

which lies within the population of 45,000 individuals. Thus, the systematic sample consists of exactly 680 individuals selected at equal intervals throughout the population.